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An Evaluation of Different Annotation Approaches for YOLOv8 Instance Segmentation in UAV Imagery: A Case Study on the UAV-BD Dataset

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Abstract. This paper presents a comprehensive analysis of the impact of distinct annotation approaches, namely bounding boxes (B-Box) and masks, on the accuracy of YOLOv8-seg, a state-of-the-art instance segmentation model, when applied to unmanned aerial vehicle (UAV) imagery. Our investigation centres on the UAV-BD dataset, a challenging and diverse dataset designed for UAV applications. Through meticulous comparisons, we assess the performance of YOLOv8-seg using mAP50 and mAP50-95 metrics under different annotation approaches, providing an understanding of the efficacy of each annotation method. Our study yields intriguing findings: while mAP50 scores exhibit comparability between bounding box and mask annotations, a distinctive divergence surfaces in the mAP50-95 metric. The mask annotation approach consistently outperforms bounding boxes, suggesting its dominance in effectively handling instances with higher Intersection over Union (IoU) thresholds under all the backgrounds (i.e., sand, lawn, bush, land, step, ground and playground). The enhanced performance of mask-based annotations, particularly at higher IoU thresholds. This research provides valuable insights that can inform decision-making in digital twin applications and other domains reliant on accurate UAV-based object detection, fostering advancements in technology and application domains.

Keywords: Annotation Approach, Instance Segmentation, Riparian Waste Detection, UAV

1 Introduction

Plastic pollution, ranging from large macro-plastic debris to microscopic micro-plastic particles, has emerged as a pervasive global environmental issue ^[1,2]. The ubiquity, persistence, and continual accumulation of plastic waste across terrestrial and aquatic ecosystems pose severe threats to wildlife through ingestion and entanglement ^[3,4]. Micro-plastics have penetrated food chains, raising concerns about potential impacts on human health ^[5,6].

This pollution disrupts critical ecosystem processes like nutrient cycling and oxygen levels ^[7,8]. The multitude of sources contributing to the influx of plastics, from consumer products to industrial activities, presents a complex challenge ^[9]. Addressing plastic pollution demands a coordinated global effort to mitigate current impacts and drive sustainable solutions to prevent further environmental degradation ^[10].

In this research, derived from the mentioned issues, the authors chose to focus on the bottle-related targets in the natural environment as the research objects with following several bullet-point reasons:

- One of the most prevalent and clear visible forms of plastic pollution in the environment.
- The buoyancy of bottles may pose ingestion hazards to wildlife.

- Bottle caps present a further micro-plastic hazard as detaching and breaking down over time.
- Branded bottles can potentially be traced back to their point sources.
- Bottle plastics like polyethylene terephthalate (PET) are designed to be extremely durable, allowing bottles to persist for long periods in the environment before fully degrading.

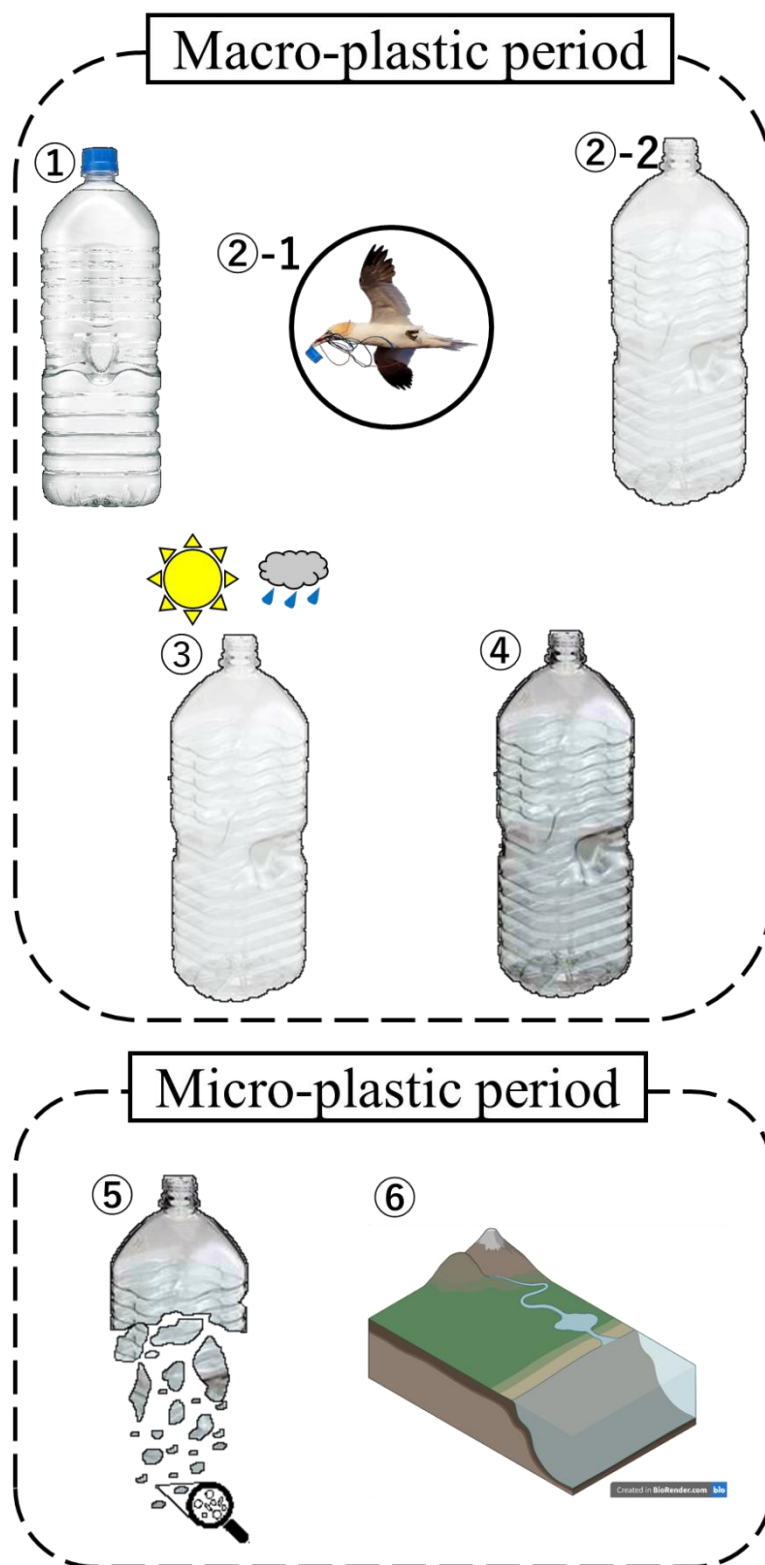


Figure 1 Processing of transferring the macro-plastic to the micro-plastic from a PET bottle sample: Step-1, PET bottles were thrown away by humans in the natural environment after drinking; Step-2, the bottle top was taken by the wild birds as the food and the bottle body still laid in the natural

environment; Step-3, after the long-term stay in the natural environment, the bottle body faced with different weather conditions (i.e., rainy, sunny days); Step-4, the color of the bottle body became darker comparing with the bottle body in Step-3 because of the sunshine and rainfall; Step-5, the plastic body started to become the separated pieces; Step-6, the separated pieces can be flowed by the rainfall or flood into the ecosystem that have negative impact on the people.

Around how the PET bottle is staying in the natural environment, Figure 1 illustrates the degradation process of plastic over time, transitioning from macro-plastic to micro-plastic. Initially, the whole plastic bottle with the blue cap was shown in step-1, which breaks down into smaller fragments and eventually micro-plastic particles that can pollute the environment, including waterways represented by the landscape images at the end. Along the way, environmental factors such as sunlight and rainfall, symbolized by the sun and cloud icons in Figure 1, contribute to this degradation process. The image depicted the long-term effects of plastic waste and its potential harm to ecosystems and wildlife.

Based on the mentioned situation, to avoid the micro-plastic coming into the natural environment because of the sunlight and rainfall, an approach that can detect the PET bottles during the macro-plastic period is in need. Considering of applying approach in the wide area, a platform like drone or Unmanned aerial vehicle (UAV) that can move flexibly is necessary. UAV imagery plays a pivotal role in diverse applications, ranging from surveillance to environmental monitoring. Accurate object detection and segmentation in UAV imagery are critical for extracting meaningful information and enhancing the capabilities of applications such as digital twin technology.

Except of considering the platform issues, annotation approach (i.e., the hints that guide the model to learn) for generating the dataset is another challenge for the researchers. Because the annotation approaches can affect the performance of the model derived from the area for training, in this paper, the authors aim to evaluate the performance of the model when using different annotation approach on the same dataset.

This paper delves into a detailed examination of the impact of different annotation methods, specifically bounding boxes (B-Box) and masks, on the accuracy of YOLOv8^[11], a cutting-edge instance segmentation/object detection model, when applied to UAV imagery. The UAV-BD dataset^[12] serves as the focal point of our investigation, providing a challenging and diverse set of scenarios that mimic real-world UAV applications.

Designed to encompass various environments, including sand, lawn, bush, land, step, ground, and playground, the UAV-BD dataset reflects the complexity of UAV imagery encountered in practical scenarios. The original annotation of the UAV-BD is totally derived from the bounding box. Instead of using the original one, as depicted in Figure 2, the authors applied the automatic annotation generator (i.e., training the model using several public datasets with plastic bottles) to segment the UAV-BD dataset to get the masks. And as shown in Figure 3, based on the minimum and maximum x-, y- position of the masks, the authors can also get the new bounding box (i.e., middle position and width/height size of the mask).

This research adopts a meticulous comparative approach, employing mean Average Precision at IoU 50 (mAP50, a measure of the accuracy of the model considering only “simple” detection) and mean Average Precision between IoU 50 and 95 (mAP50-95, provides a comprehensive view of the model’s performance at different detection-difficulty levels) metrics to assess the performance of YOLOv8 under distinct annotation approaches (i.e., B-Box and mask derived from the automatic annotation generator).

One of the assessment metric index in YOLOv8, mAP50/50-95 (B) refers to the performance of the bounding box in object detection. In Figure 3, a true positive is when the IoU between the predicted bounding box and the ground truth is above a certain threshold (0.5 in mAP50, 0.5~0.95 in mAP50-95), and a false positive is when it is below.

And mAP50/50-95 (M) is the equivalent of mask IoU, which works differently. Besides bounding boxes and class labels, YOLOv8 predicts segmentation masks for each object. These represent the pixel-level regions corresponding to the class of the object. The mask IoU is the measure of the pixel-wise overlap between the predicted mask and the ground truth mask shown in Figure 3, and applies only to the mask within each predicted object bounding box. If the mask IoU exceeds a certain threshold (typically 0.5, 0.5~0.95), the authors consider this a true positive, meaning that the segmentation of the model matches the ground truth well. If it falls below the threshold, it's considered a false positive.

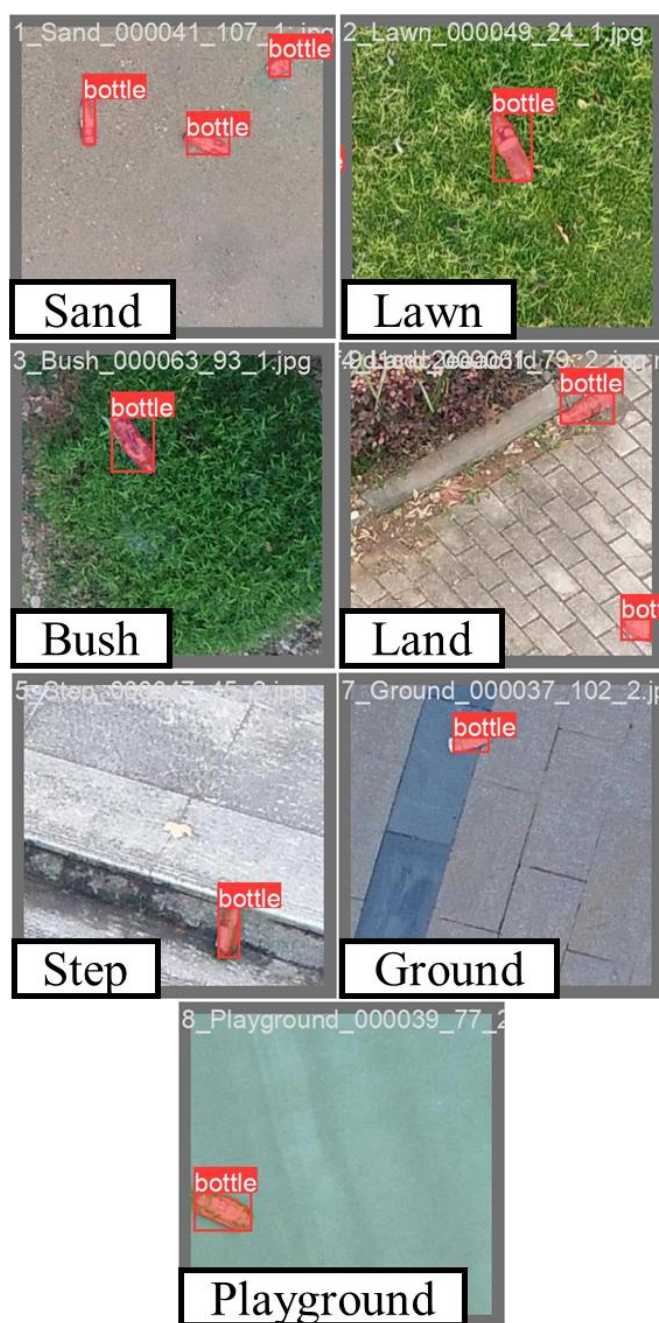


Figure 2 True Label (TL) and evaluation approach difference.

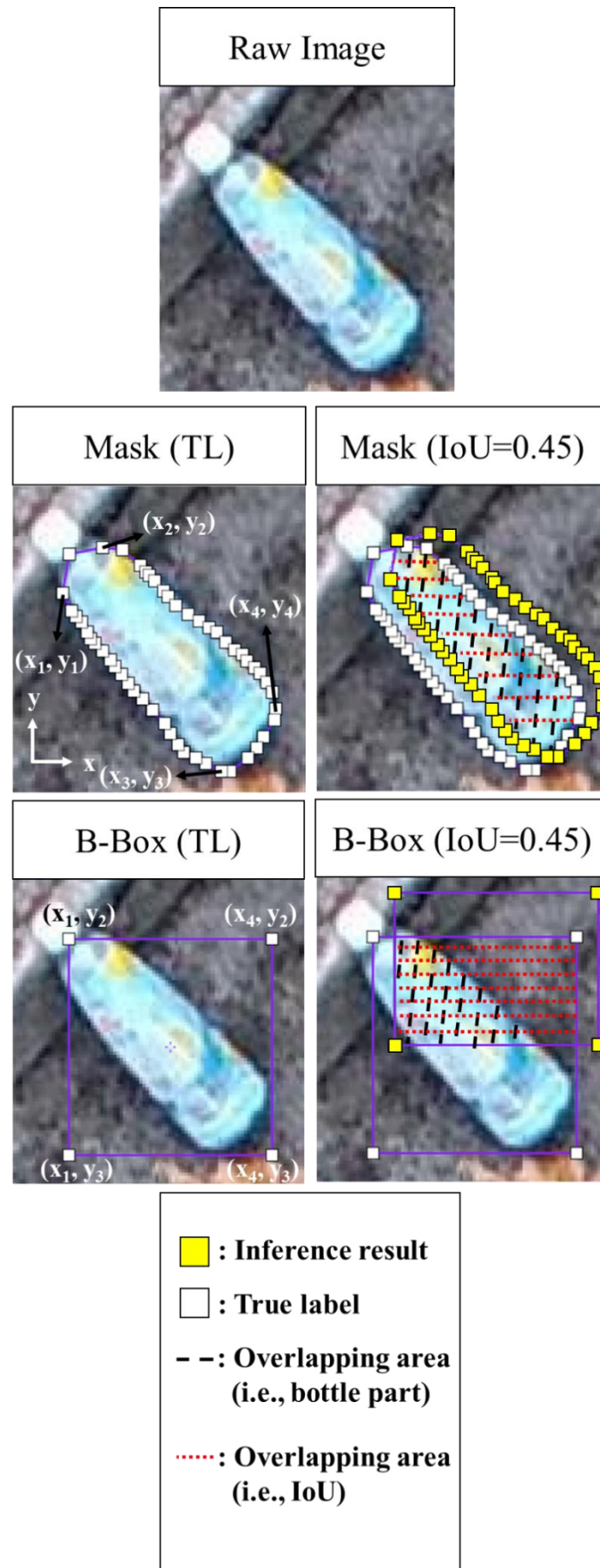


Figure 3 True Label (TL) and evaluation approach difference.

Table 1 Image numbers for Train/Valid/Test in individual group.

Groups	Image numbers		
	Train	Valid	Test
1_Sand			
2_Lawn			
3_Bush			
4_Land	827	207	258
5_Step			
7_Ground			
8_Playground			
0_All (Total)	5789	1449	1806

2 Work flow

Training UAV-BD dataset (train/valid) as shown in Table 1, with generated B-Box and mask, inferring the UAV-BD dataset (test part) and getting the results derived from both the mask- and B-Box- based standards (i.e., True Label and evaluation approach). To understand the accuracy derived from the annotation difference, the authors compared the Precision, Recall, mAP50, mAP50-95 and F1 score (IoU=0.45) between mentioned different two annotation approach (i.e., mask-based value minus bounding-box-based value) under various backgrounds.

3 Conclusion

As shown in the Figure 4, the findings of our study reveal intriguing nuances in the performance of YOLOv8-seg under different annotation approaches. Notably, while mAP50 scores exhibit comparability between bounding box and mask annotations, a distinct divergence emerges in the mAP50-95 metric. The mask annotation approach consistently outperforms bounding boxes, showcasing its dominance in effectively handling instances with higher IoU thresholds across various backgrounds.

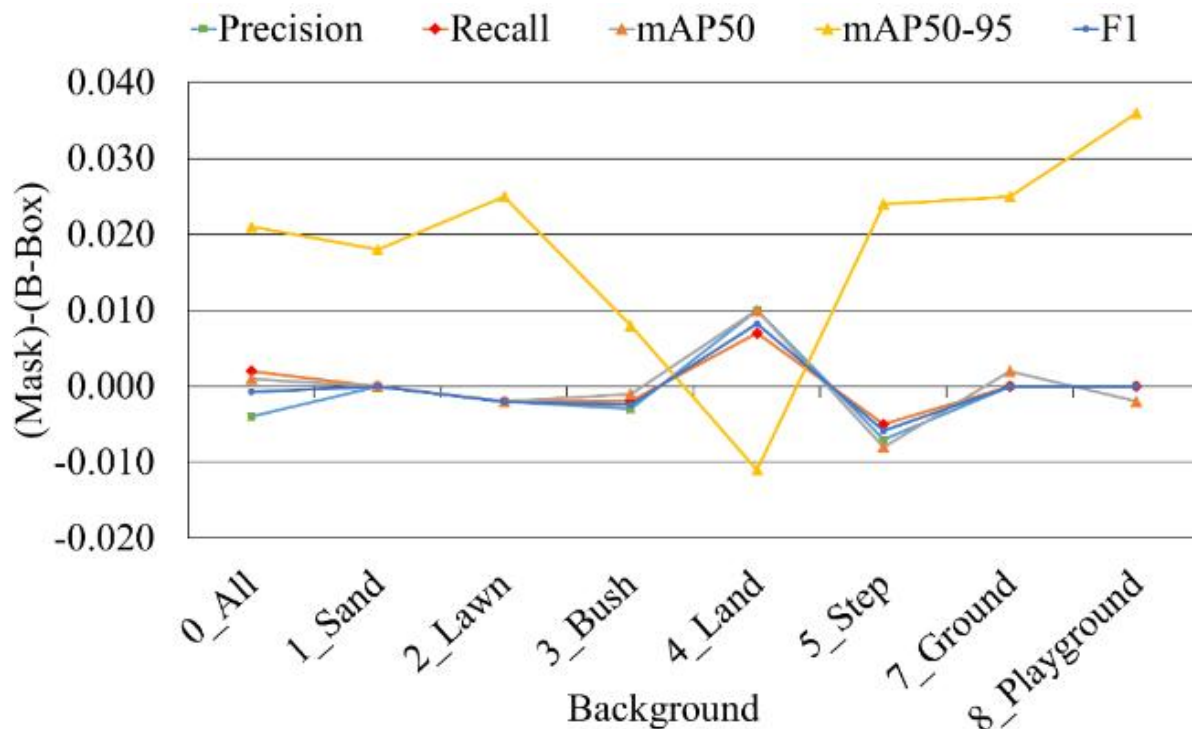


Figure 4 Comparison of the F1 score (IoU=0.45), mAP50 and mAP50-95 between the mask- and bounding-box- based standards.



Figure 5 A sample result in 4_Land group that include two overlapping bottles with similar colors.

4 Discussion

Considering the reasons of the evaluation-index-based difference, Figure 3 has depicted that the backgrounds in the bounding box have been trained in the model, the bottle-based overlapping area probably have been reduced. And as depicted Figure 5, the bottles were overlapped by each other which is difficult for the trained model to segment one from another, especially the two bottles have the similar color. It is also another reason that effected the accuracy in 4_Land group.

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